

Team FraudShield

FraudShield AI

Adaptive Fraud Intelligence Engine with Macro-Surge & Temporal Velocity Decoupling

ALLOW REVIEW BLOCK

The flash sale dilemma

Static rules panic

Fast purchases in a short window get flagged, whoever is buying.

Real buyers get blocked

On Diwali sale thousands check out at once. Rules cannot tell a crowd from a crime.

Bots hide in the rush

Fraud rings blend into peak traffic or fire rapid micro-payment bursts.

Traditional system: real buyers blocked

Flash sale crowd → **Speed rule fires** → **Real buyers blocked**

Target goal: bots separated

Flash sale crowd → **Check who is behind the speed** → **Bots stopped, buyers pass**

What the data tells us

Raw user data
IP, device, location

→

Anonymization layer
28 protected features, V1 to V28

→

Key risk signals
V14 and V17 flag unusual behavior

Privacy first

Personal details are compressed away before any analysis happens.

Two strong signals

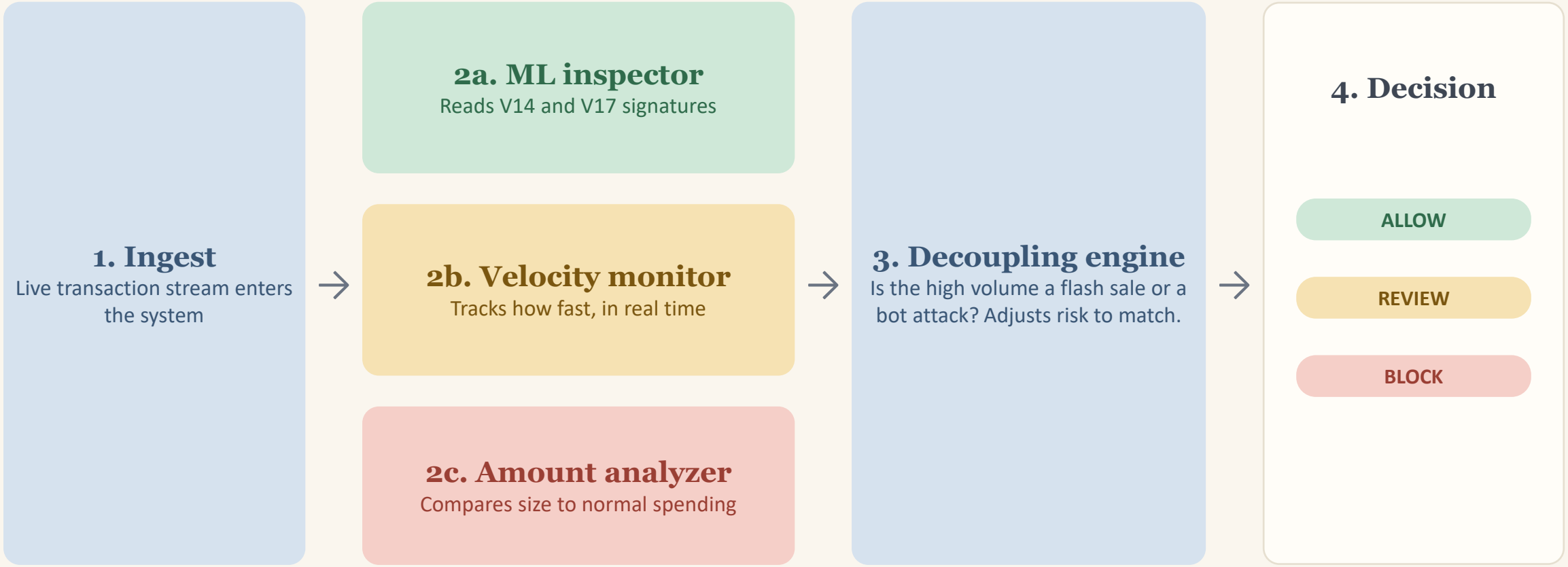
V14 and V17 point to abnormal user behavior most often.

Time and amount

Transaction time and value reveal sudden spending spikes.

How FraudShield AI works

Runs three checks in parallel



The core intelligence, in plain English

XGBoost

The master inspector

Weighs many faint clues together to judge fraud likelihood.

Like a detective connecting small details.

PCA

The privacy shield

Removes personal identity, keeps the risk pattern.

Like blurring a face but keeping the body language.

Z-score

Spending anomaly meter

Shows how far a purchase strays from usual spending.

A coffee-a-day shopper suddenly buys a laptop.

Time-decay

The speed tracker

Counts recent bursts more than old activity.

Ten taps in one second beats ten in an hour.

The secret weapon: adaptive surge decoupling

High transaction speed detected: now check the feature signals

Flash sale path

High velocity + normal signals

↓

Legitimate surge: velocity penalty lowered

↓

ALLOW: real buyers pass smoothly

Botnet attack path

High velocity + suspicious signals

↓

Coordinated fraud: risk escalated

↓

BLOCK: bots are stopped

One score, three actions

A composite 0 to 100 score blends ML probability, surge-adjusted velocity, amount anomaly and the surge indicator

0 to 40: Low risk	40 to 70: Medium	70 to 100: High

ALLOW

Instant approval.

REVIEW

One-time passcode or manual review queue.

BLOCK

Automatic rejection.

Interactive dashboard prototype

Live stream simulator

Normal traffic	ALLOW
Flash sale surge	ALLOW
Coordinated bot attack	BLOCK
Unusual single purchase	REVIEW

Transaction inspector

Click any row for its risk breakdown and a plain-English explanation.

Dynamic calibration

Review from



Block from



Risk managers move thresholds live.

Impact and business value

Up to 35%

fewer false declines during flash sales

Under 50 ms

real-time evaluation of streaming transactions

100%

PII compliant, zero compromise on privacy

Less

manual review workload for the risk team